10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 11.1 — Information Handling (Graphs of Frequency Distributions)

Unit 11 | Exercise 11.1 | New Book (PECTAA)

Exercise 11.1 (Solved)

This exercise is all about reading information from a graph: an ogive for the median, quartiles, deciles and percentiles, and a box-and-whisker plot or a scatter diagram for the spread and the trend of the data.

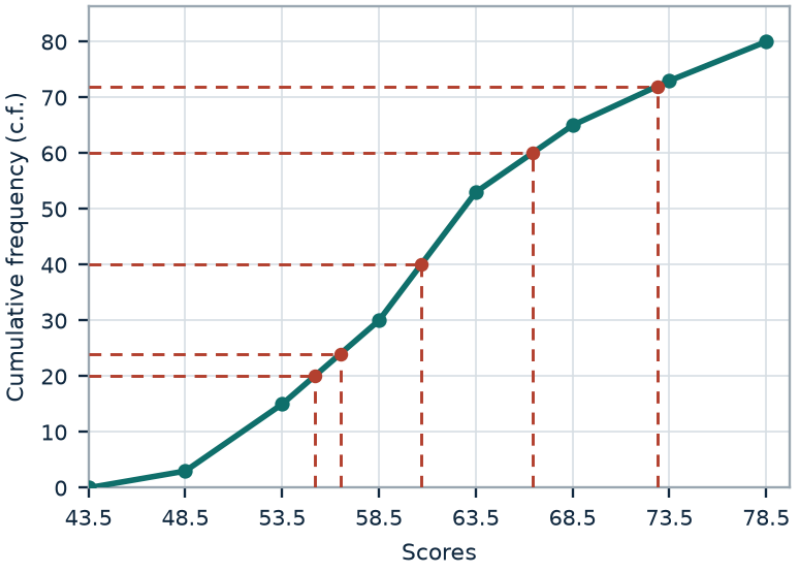
1. The following frequency distribution represents the achievement of a student's group upon a memory test. (i) Plot cumulative frequency graph of group scores. (ii) Determine median, lower and upper quartiles, D_3, P_{90} and interquartile range graphically. (Note: write your answers in whole numbers.)

Table of cumulative frequencies

Scores	No. of students (f)	Cumulative frequency (c.f.)
44 – 48	3	3
49 – 53	12	3 + 12 = 15
54 – 58	15	15 + 15 = 30
59 – 63	23	30 + 23 = 53
64 – 68	12	53 + 12 = 65
69 – 73	8	65 + 8 = 73
74 – 78	7	73 + 7 = 80

The upper class boundaries are 48.5, 53.5, 58.5, 63.5, 68.5, 73.5 and 78.5, and the curve starts from the point (43.5, 0).

Here n = Σf = 80.



Cumulative frequency polygon (ogive) of the memory test scores

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Position of median = n / 2 = 80 / 2 = 40

Position of Q_1 = n / 4 = 80 / 4 = 20

Position of Q_3 = 3n / 4 = 3(80) / 4 = 60

Position of D_3 = 3n / 10 = 3(80) / 10 = 24

Position of P_{90} = 90n / 100 = 90(80) / 100 = 72

Now we draw a horizontal line from each of these positions on the y-axis to the curve, and then a vertical line down to the x-axis. The reading on the x-axis is the required value.

Readings taken from the graph

Measure	Position on y-axis	Reading on x-axis
Median	40	61
Q_1 (lower quartile)	20	55
Q_3 (upper quartile)	60	66
D_3	24	57
P_{90}	72	73

Interquartile range = $Q_3 - Q_1$ = 66 − 55 = 11

Answer: Median = 61, Q_1 = 55, Q_3 = 66, D_3 = 57, P_{90} = 73, IQR = 11

2. Construct an ogive for the following distribution of scores. Determine median, lower and upper quartiles, D_6, P_{80} and interquartile range graphically.

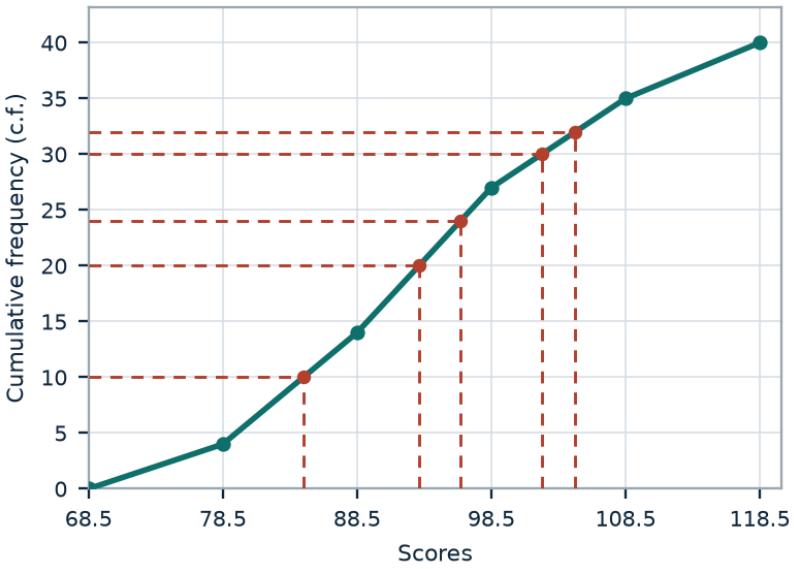
Table of cumulative frequencies

Scores	f	Cumulative frequency (c.f.)
69 – 79	4	4
79 – 89	10	4 + 10 = 14
89 – 99	13	14 + 13 = 27
99 – 109	8	27 + 8 = 35
109 – 119	5	35 + 5 = 40

The upper class boundaries are 78.5, 88.5, 98.5, 108.5 and 118.5, and the curve starts from the point (68.5, 0).

Here n = Σf = 40.

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Cumulative frequency polygon (ogive) of the given scores

Position of median = n / 2 = 40 / 2 = 20

Position of Q_1 = n / 4 = 40 / 4 = 10

Position of Q_3 = 3n / 4 = 3(40) / 4 = 30

Position of D_6 = 6n / 10 = 6(40) / 10 = 24

Position of P_{80} = 80n / 100 = 80(40) / 100 = 32

Readings taken from the graph

Measure	Position on y-axis	Reading on x-axis
Median	20	93.1
Q_1 (lower quartile)	10	84.5
Q_3 (upper quartile)	30	102.25
D_6	24	96.2
P_{80}	32	104.75

Interquartile range = $Q_3 - Q_1$ = 102.25 − 84.5 = 17.75

Answer: Median = 93.1, Q_1 = 84.5, Q_3 = 102.25, D_6 = 96.2, P_{80} = 104.75, IQR = 17.75

3. Find Q_1, Q_3, median, range and IQR for the dataset given below: 3, 6, 8, 4, 7, 5, 10, 11, 13, 9, 14, 12, 15, 16, 17. Also draw a box-and-whisker plot.

First we arrange the data in ascending order.

3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17

Here n = 15, so the median is the 8th value.

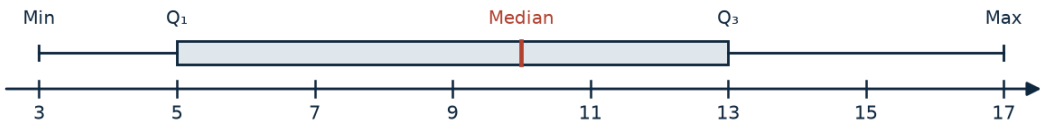
Median = 8th value = 10

Q_1 = median of the first seven values = 5

Q_3 = median of the last seven values = 13

Range = highest value − lowest value = 17 − 3 = 14

IQR = $Q_3 - Q_1$ = 13 − 5 = 8



Box-and-whisker plot: Min 3, Q_1 5, Median 10, Q_3 13, Max 17

Answer: $Q_1 = 5$, $Q_3 = 13$, median = 10, range = 14, IQR = 8; box plot drawn

4. Find Q_1, Q_3, median, range, IQR and extreme values for the following data: 102, 98, 95, 100, 93, 110, 108, 104, 97, 96, 92, 101, 99, 105, 107. Also draw a box-and-whisker plot.

Arranging in ascending order.

92, 93, 95, 96, 97, 98, 99, 100,

101, 102, 104, 105, 107, 108, 110

Here n = 15, so the median is the 8th value.

Median = 8th value = 100

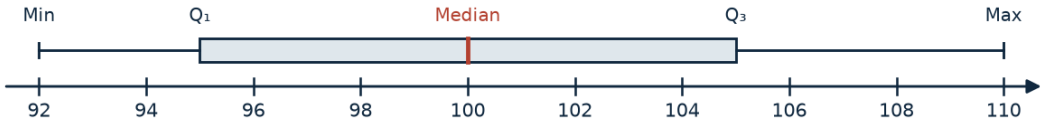
Q_1 = median of the first seven values = 95

Q_3 = median of the last seven values = 105

Range = 110 − 92 = 18

IQR = $Q_3 - Q_1$ = 105 − 95 = 10

Extreme values: minimum = 92, maximum = 110



Box-and-whisker plot: Min 92, Q_1 95, Median 100, Q_3 105, Max 110

Answer: $Q_1 = 95$, $Q_3 = 105$, median = 100, range = 18, IQR = 10, min = 92, max = 110

5. Find Q_1, Q_3, median, minimum and maximum values for the following box-and-whisker plot:



Given box-and-whisker plot (scale: one small division = 2 units)

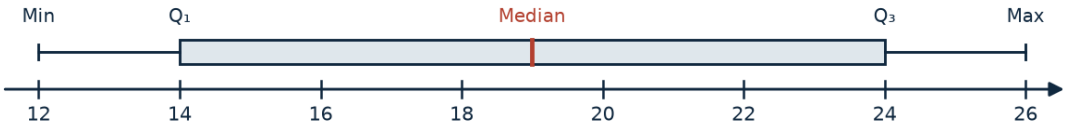
Reading the plot against the scale on the number line.

Minimum = 10, Q_1 = 22, median = 25, Q_3 = 35, maximum = 46

Answer: Minimum = 10, $Q_1 = 22$, median = 25, $Q_3 = 35$, maximum = 46

6. Draw the box-and-whisker plot, if min. value = 12, max. value = 26, median = 19, Q_1 = 14 and Q_3 = 24.

All five values are given, so we mark them on a number line and join them.



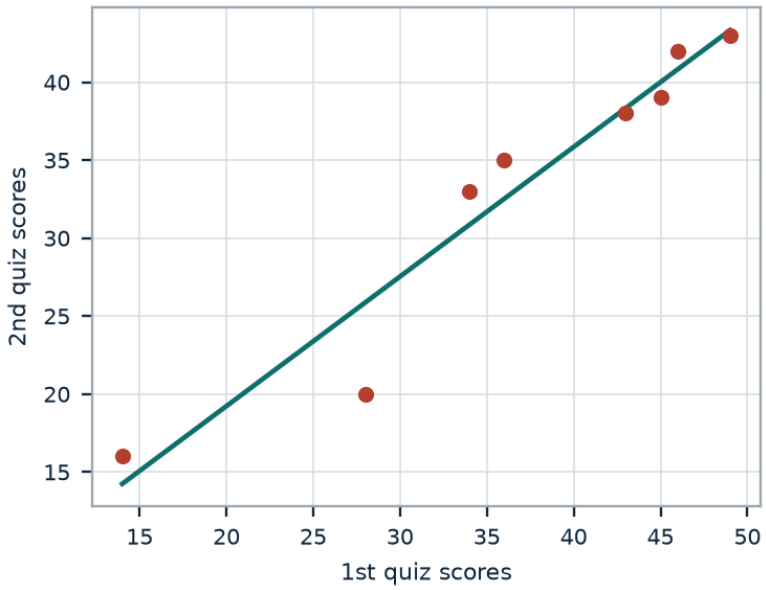
Required box-and-whisker plot

Answer: Box plot drawn with Min 12, Q_1 14, Median 19, Q_3 24, Max 26

7. Construct a scatter diagram and draw a line of best fit for the following quiz scores for 8 students in a class. Describe correlation between 1st and 2nd quiz scores also.

Quiz scores of 8 students

1st quiz	14	28	34	36	43	45	46	49
2nd quiz	16	20	33	35	38	39	42	43



Scatter diagram with the line of best fit

Each pair is plotted as one point. The points rise from left to right and lie close to a straight line, so the correlation is strong and positive.

Answer: Graph drawn; the correlation between the 1st and 2nd quiz scores is strong positive

Remember

Median at n / 2, Q_1 at n / 4, Q_3 at 3n / 4 on the c.f. axis.

IQR = $Q_3 - Q_1$; in a scatter diagram, rising points = positive correlation.